YUANKAI TAO

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EDUCATION

Georgia Institute of Technology *Aug 2026 — Dec 2027*
M.S. in Operations Research, Stewart School of Industrial and Systems Engineering
Selected Coursework: Deterministic Optimization, Computational Statistics, Simulation

University of California, San Diego *Sep 2024 — Dec 2025*
M.S. in Quantitative Finance, Rady School of Management *GPA:* 3.84/4.0
Selected Coursework: Data Science for Finance, Deep Learning, Computational Finance Methods

Zhongnan University of Economics and Law *Sep 2020 — Jun 2024*
B.S. in Management, School of Accounting and Finance *GPA:* 3.63/4.0
Selected Coursework: Probability & Statistics, Linear Algebra, Calculus, Python Programming

TECHNICAL SKILLS

Programming & Querying **Python, SQL**, MATLAB, STATA, VBA, LaTeX
Data & Cloud **PostgreSQL, PostGIS, BigQuery, DuckDB, GCP**, Linux, Git
ML & Analytics Statistical Modeling, Time Series, Feature Engineering, Model Selection, Validation
Libraries & Tools **Pandas, Polars, NumPy, scikit-learn**, PyTorch, Matplotlib, LightGBM, XGBoost

PROFESSIONAL EXPERIENCE

VariFlight Technology *Jul 2025 — Sep 2025*
Data Analyst, Intern *Data Development Department | Anhui, China*

• Built Python web-scraping workflows to collect public reports, webpages, and images from the Civil Aviation Administration of China for recurring data analysis.

• Applied OCR and table-extraction tools to transform **600+ unstructured PDF reports and 500+ webpages** into standardized, analysis-ready datasets for local LLM training; added repeatable cleaning and quality-control steps.

• Created dashboards for transportation and passenger-turnover metrics, visualizing operational trends and enabling faster interpretation of key aviation indicators.

Guangdong Hedge Fund Investment Management Co., Ltd. *May 2026 — Jul 2026*
Quantitative Research, Intern *Quantitative Strategy Department | Hong Kong*

• Developed an end-to-end ML pipeline with 16 trend, momentum, volatility, liquidity, volume, and open-interest features; trained LightGBM, XGBoost, CatBoost, and neural-network models across 1- to 20-day horizons.

• Designed rolling, time-aware validation and retraining workflows; evaluated models with statistical, risk, and transaction-cost metrics to reduce leakage and support reproducible comparison.

• Implemented a monthly rebalanced model in Backtrader that achieved a **1.89 Sharpe ratio**, **16.79% annualized return**, and **25.51% excess return** after transaction costs.

Ernst & Young *Dec 2022 — Feb 2023*
Audit, Intern *Financial Service Office (FSO) | Beijing, China*

• Automated the cleaning and consolidation of financial tables from **220+ CSV files**, generating standardized audit outputs and improving the consistency of recurring data-review workflows.

• Performed variance analysis on intercompany and investment transactions, investigated anomalies, and validated findings against supporting records.

SELECTED PROJECTS

Kaggle Jane Street Market Prediction Competition *Silver Medal*

• Built a memory-efficient Polars/PyTorch pipeline for **47M intraday observations** across 1,669 trading days; engineered lagged, rolling, and temporal features and deployed online inference through Kaggle's streaming evaluation API.

• Implemented a parameter-efficient TabM regressor with a **32-member shared-parameter ensemble**; one-hot encoded three categorical features and improved cross-validation performance by **0.004**.

• Contributed to the baseline training and final ensemble of XGBoost, MLP, Ridge, and TabM, helping the team place **94th of 3,757 teams**.

Text Analysis and BERT Fine-tuning on SEC 10-K Reports *MGTF 423, UCSD*

Team Lead | Analysis and Human-in-the-Loop Learning on 20.5M Text Rows

• Performed named-entity recognition, frequency analysis, and visualization on SEC filings to identify patterns in corporate disclosures.

• Built a human-in-the-loop labeling and fine-tuning workflow with three annotation rounds, achieving **79% accuracy** in classifying uncertainty tone under limited labeled data.

UCSD / FMSbonds, Inc. 2025 Capstone Project *Aug 2025 — Dec 2025*

Team Lead | High-Performance Bond Pricing Engine Optimization

• Designed caching for high-frequency cash-flow queries, eliminating redundant object generation and reducing per-bond runtime from 1.17 ms to 0.13 ms (**9× speedup**).

• Reimplemented the yield-to-worst solver with Numba JIT, reducing runtime from 1.57 ms to 7.5 μs per bond (**210× speedup**) for scalable batch analytics.

RESEARCH EXPERIENCE

Research Assistant for Professor Franklin Qian, UNC-Chapel Hill *Mar 2025 — Jan 2026*

The Numbers Game: Signaling Effects of Pricing Formats in Housing Bargaining

• Engineered an end-to-end Python and SQL pipeline for a **76-million-observation spatial dataset** with 40+ variables on Google Cloud Platform, integrating property and population data from multiple sources.

• Developed parallelized PostGIS workflows with property-specific filters across multiple distance ranges, achieving approximately **50× computational speedup** while preserving reproducible analytical logic.

• Built and managed a **964.5M-record, 100GB+ database** using PostgreSQL, BigQuery, and DuckDB on GCP and UNC Longleaf HPC; executed complex geospatial and time-window queries for downstream research.

CERTIFICATIONS & HONORS

Deep Learning Specialization (5-Course Certificate) — Coursera, DeepLearning.AI

Algorithms and Data Structures Certificate (6-Course Certificate) — Coursera, UC San Diego

Beta Gamma Sigma (Top 20% of graduating class) — UCSD, *2024 — 2025*

WorldQuant Alphathon 2024, Gold Level (Rank 16, Top 3.1%) — *2024*

Special Scholarship in Innovation and Entrepreneurship — ZUEL, *2022 — 2023*